INCH-POUND

MIL-T-10010J
28 June 1994
Superseding
MIL-T-10010H
28 June 1989

MILITARY SPECIFICATION

TRACTOR, FOUR WHEELED, SIT DOWN, WAREHOUSE, ELECTRIC-MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tractors, four wheeled, sit down, warehouse, electric motor driven, solid rubber tires, used for pulling loaded or unloaded trailers in warehouses or on improved surfaces.

1.2 Classification. Tractors shall comply to the following parameters:

1.2.1 Type. (see 6.2)

a. Type E - Non-spark enclosed

b. Type EE - Spark enclosed

1.2.2 Maximum Drawbar Pull (DBP). (see 6.2)

a. Size I 2,000 lbs (909 Kg)

b. Size II 3,000 lbs (1364 Kg)

c. Size III 4,000 lbs (1818 Kg)

d. Size IV As specified

Beneficial comments (recommendations, additions, deletions) and any other pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Navy Ships Parts Control Center, Code 0341 P.O. Box 2020 , Mechanicsburg, PA 17055-0788 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation, (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-133	- Batteries, Storage (Lead Acid, Industrial for Cycle Service)
GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum)
QQ-P-416	- Plating, Cadmium (Electrodeposited)
TT-E-489	- Enamel, Alkyd, Gloss, (for Exterior and Interior Surfaces)

MILITARY

MIL-P-514	- Plate, Identification, Instruction and Marking, Blank
MIL-T-704	- Treatment and Painting of Material
MIL-F-3541	- Fittings, Lubrication
MIL-G-3859	- Grease Gun, Hand, High Pressure, Lever-Operated, Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity)
MIL-M-3971	- Meter, Time Totalizing, Non-Hermetically Sealed, Electrical, General Specification for
MIL-T-17479	- Trailer, Platform, Warehouse, Wood or Metal Platform, 6000 Pound Capacity, Pneumatic Tires, Fifth Wheel Steer

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STANDARDS

FEDERAL

FED-STD-595 - Colors

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-162 - Materials Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routing, Testing and Processing

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility

MS-15367 - Battery, Storage, for Electric Powered Industrial Trucks and Tractors

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government Documents. The following other Government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

U.S. Department of Labor

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards (OSHA)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS (ASTM Standards)

ASTM B633 - Electrodeposited Coating of Zinc on Iron or Steel

ASTM D3951- Packaging, Commercial

(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION STANDARDS (NEMA Standards)

NEMA MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington D.C. 20037)

SOCIETY OF AUTOMOTIVE ENGINEERS HANDBOOK (SAE Handbook)

SAE J154 - Operator Enclosures Human Factor Design Considerations

SAE J209 - Instrument Face Design and Location for Construction and Industrial Equipment

SAE J297 - Operator Controls on Industrial Equipment

SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz)

SAE J588 - Turn Signal Lamps

SAE J589 - Turn Signal Switch

SAE J833 - USA Human Physical Dimensions

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SAE J898 - Control Locations for Construction and Industrial Equipment Design

SAE J899 - Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines

SAE J925 - Minimum Access Dimensions for Construction and Industrial Machinery

SAE J985 - Vision Factors Considerations in Rear View Mirror Design

SAE J994 - Alarm-Backup-Electric, Performance, Test and Application

SAE J1362 - Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) (Current Issues)

ANSI B56.9 - Standard of Safety for Controlled Industrial Tow Tractor

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018)

THE TIRE AND RIM ASSOCIATION, INC. YEARBOOK (T & RA Yearbook)

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market St., Akron, OH 44313)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 583 - Standard for Electric - Battery - Powered Industrial Trucks

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York 11756; 207 East Ohio Street, Chicago, Illinois 60611 or 1655 Scott Boulevard, Santa Clara, CA 95050)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified a first produced unit shall be subjected to a first article inspection (see 6.2) in accordance with 4.4.

3.2 General. The tractor shall be electric motor driven, sit down, warehouse, rear-wheel drive, front wheel steer, solid rubber tired, four wheel sprung suspension, and four wheel split brake system. Except for batteries, the tractor shall be constructed and equipped to meet all requirements specified herein and shall be complete and ready for operation when delivered. Batteries shall be as specified in 3.7.3. The end items and major and minor assemblies shall be new and of a standard design or model product supplied by the end item manufacturer current at time of bidding, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. There shall be no change in components to offset interchangeability within individual contract items without authorization by the contracting officer. The component parts of the unit need not be the products of the same manufacturer; however, the manufacturers shall be so established in the industry that prompt and continuing service and delivery of replacement parts may be assured.

3.3 Materials. Materials not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet test requirements specified in Section 4. "Normal commercial product" shall be interpreted to mean an item furnished by an end item manufacturer normally engaged in commercial production of the item(s) covered by this specification. Asbestos material shall not be used in the manufacture of any part of this vehicle.

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3.3.1 Castings. Castings shall be of the quality normally used in commercial practice, free from blowholes, porosity, shrinkage, cracks, or other defects. Castings shall not be repaired in a matter that adversely affects the strength or intended use. Sufficient finish shall be allowed to permit machining for leak-free sealing at covers, closure points, and fasteners.

3.3.2 Plating. All threaded fasteners, washers and cotter pins required to fabricate the truck, except those in contact with oils in reservoirs and those inside corrosion resistant components shall be zinc plated, cadmium plated or made of corrosion resisting material. Zinc plating shall be in accordance with ASTM B633, type I, SC 1 and cadmium plating shall be in accordance with QQ-P-416, type I, class 3.

3.3.3 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.4 Safety. The truck shall conform to the safety standards issued by the Occupational Safety and Health Administration (OSHA) under the Secretary of Labor, and to the applicable requirements in the current issue of UL583 and ANSI B56.9. The contractor shall have one tractor tested by an independent testing laboratory to certify that the design of the tractors meets or exceeds the requirements of ANSI/UL.

3.5 Recycled, Reclaimed, and Virgin Materials. Recycled and reclaimed materials shall be used to the maximum extent possible without hindering the intended use or without weakening any part of the tractor. There is no requirement that the tractor be manufactured from virgin materials.

3.6 Maintainability. Provision shall be made for ready adjustment, servicing, or replacement of all electrical assemblies and components, battery, brake and components, wearing parts of steering assembly, tires, wheels, and horn without the removal of any major parts other than covers. Hand access openings used on the exterior of the truck must conform to the requirements of SAE J925. The edge of the opening shall be smooth and shall be provided with a removable or hinged cover. Frame members or other obstructions shall not interfere with the servicing of or draining of oils and lubricants from any assembly or components. Servicing or draining shall not require the use of flexible connections or other special device. Drain plugs shall provide for complete drainage. Compliance with specific maintenance requirements for the equipment shall be determined in accordance with 3.20.9 and 4.6.12.

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3.7 Design . The design of the tractors and all components covered by this specification shall provide safe and efficient means for pulling trailers unloaded or loaded with palletized and unpalletized loads of the rated capacity over paved areas and in warehouses. There shall be no evidence of accelerated wear, or failures, or permanent deformation when operated under normal service conditions or when tested as specified in Section 4.

3.7.1 Electrical equipment.

3.7.1.1 Resistors. Resistors shall not be used in series with the motor armature for speed control. Speed control shall be effected by the application of a system as specified in 3.7.2.4. Low resistance shunts across field coils may be used to increase top speed of the motor. Such resistors as are necessary in the control circuit network shall be continuous-duty types of a power rating value sufficient to withstand all tests required by this specification.

3.7.1.2 Enclosures and Shields. All tractors shall comply with the requirements of UL 583. All electrical components including wiring shall be located or protected by adequate means to prevent contact by excess moisture or battery electrolyte in the event of spillage.

3.7.1.3 Overcurrent devices. In addition to the requirements of 3.4, overcurrent devices shall be provided to properly protect each power circuit from damage resulting from wiring defects or other faulty operations. The overcurrent devices shall not disconnect power when specified tests, including slope ascension, are conducted with fully charged batteries.

3.7.1.4 Heat Producing Devices, Type EE Tractors. For EE rated tractors, the outside surface temperature of heat-producing devices shall stabilize at temperatures which are allowed by UL 583 while the tractor is operating with rated capacity continuously at the speed which creates the most heat.

3.7.1.5 Static discharge device. Type EE tractors shall be equipped with a static discharge device or static discharge type tires conforming to UL 583.

3.7.1.6 Control panel. The control panel shall be designed for easy removal as a unit or front removal of electrical assemblies mounted thereon. Mounting studs, bolts, or nuts, shall be so secured that they will not turn or become loose when assembly is removed from the panel. The control panel cover design shall permit unobstructed removal of the cover for inspection and servicing.

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3.7.1.7 Conductors. Conductors and enclosures shall comply with all requirements of UL 583. All wiring shall be color or number coded and identified to the wiring diagram.

3.7.2 Motor and motor accessories.

3.7.2.1 Motors - Type E tractors. When specified (see 6.2), the tractors shall be type E rated conforming to the requirements of UL 583 and ANSI B56.9 (see 3.4). Traction motors shall be industrial type, with a high starting torque, designed for efficient operation on 36 volts D. C. Motors shall withstand the stresses and current loads induced by test as required by this specification. The resultant strains and temperature rise shall not cause failure of any component part or excessive pitting other than normal discoloration of the commutators. The steering pump motor, if used, shall successfully withstand the stresses and current loads induced by tests required by this specification. All electric motors shall have bolt-on conductor connections.

3.7.2.2 Motors - Type EE tractors. When specified (see 6.2), the tractors shall be type EE rated conforming to the requirements of UL 583 and ANSI B56.9 (see 3.4). In addition to the requirements of 3.7.2.1, motors for type EE trucks shall be provided with one or more thermal relay switches which cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceed limits as specified in 3.7.1.4. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the over-heated circuit.

3.7.2.3 Insulation. Insulation for motors shall be Class B, F, or H as defined in NEMA MG 1, but the insulation is not to contain silicone materials. Silicone material in any form shall not be used in the motor. The electrical and mechanical properties of the insulated windings shall not be impaired by the application of the temperature permitted for the class insulation used (the word "impaired" is here used in the sense of causing any change which could disqualify the insulating material for continuous service).

3.7.2.4 Speed controller. The speed controller shall be a solid state type Silicon Controlled Rectifier (SCR) PMC or MOSFET type. The SCR shall provide infinite variation in travel speed from zero to maximum tractor speed. The speed controller shall provide smooth acceleration in both forward and reverse, and shall allow for smooth transition when reversing directions with the tractor in motion (plugging). Resistors shall not be used for speed control of the tractor. Contactors shall have replaceable tips of silver or copper alloy composition.

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3.7.2.5 Safety cutout system.

3.7.2.5.1 Type E and EE tractor. Unless otherwise specified (see 6.2) type E and EE tractors shall be equipped with a seat activated switch which interrupts all power to the traction motor when the operator leaves his seat.

3.7.2.5.2 Type EE tractors. In order to insure that no current is flowing when the connector is being disconnected on type EE tractors, a magnetic contactor shall be installed in one of the two main power lines from the battery connector. The control circuit for the magnetic contactor shall be wired with two switches in series, either of which shall cause the contactor to open and disconnect all power. The first switch shall be a seat switch. The second switch shall be an emergency switch with a positively attached mushroom type button located within easy reach of the operator. If specified (see 6.2) the seat switch shall be deleted.

3.7.3 Battery and battery accessories. Unless otherwise specified (see 6.2), battery size and capacity for type E and EE tractors shall be governed by MS-15367. The battery shall conform to W-B-133 (see 6.2). When a first produced tractor is required (see 6.2), the battery for the first produced tractor used for testing by the contractor shall be shipped in a wet and charged condition after the exterior of the battery has been rehabilitated to a "like new" appearance. When specified (see 6.2) a single point watering system shall be furnished.

3.7.3.1 Battery connector mounting. Type E and type EE tractors shall be equipped with connectors, suitably located to fit the locking half as specified in 3.7.3. For type EE tractors, provision shall be made for locking coupled connector with a padlock. Contractor does not provide the padlock.

3.7.3.2 Battery compartment. The battery compartment shall be located within the planned outline of the tractor. The compartment shall be constructed to permit quick inspection and servicing of battery using common hand tools and shall provide for easy battery removal. The inside of the compartment shall be finished with one coat of acid and alkaline resistant paint to resist deterioration through normal contact with electrolyte. Compartment construction shall provide for support to give a minimum of 1/2 inch (1.3 cm) clearance between the battery and tractor.

3.7.4 Drive unit. Each gear (with the exception of worm gears), sprocket, shaft, and axle, shall be of heat treated alloy steel.

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The drive motor(s) and rear axle shall be a single integral unit directly connected together, totally enclosed for EE tractors and operating in a constant oil bath. Drive gear housing(s) shall be equipped with filler and magnetic drain plugs. All required drain plugs, check plugs, or oil level checking and filling devices in the drive system shall be readily accessible without removing major components. Provisioning shall be made for adjustment where necessary to compensate for wear in the differential and pinion.

3.8 Construction.

3.8.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel and steel castings, and shall withstand all testing in Section 4 without permanent deformation. Frame members, bracings, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford protection for working parts. The frame and related structures shall have an adequate suspension system for reducing load shocks encountered during normal operation so that the operator can safely operate the vehicle at all times.

3.8.2 Bearings. Rotating parts shall be mounted on ball or tapered roller bearings, except as specifically indicated herein. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under normal tractor operation.

3.8.3 Wheels.

3.8.3.1 Wheels for solid tires. When solid tires are specified (see 6.2), the tractor shall be mounted on steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings. Wheel hub bearings shall provide for adjustment for proper running tolerances and shall be sealed with a retainer to prevent leakage of lubricant. Front steer wheels shall track true in accordance with automotive practices, as determined by their respective radii. Wheels, equipped with new tires, shall clear any part of the tractor structure by at least 0.25 inch (6 mm) under all conditions of operation.

3.8.3.2 Wheels for pneumatic tires. When pneumatic tires are specified (see 6.2), the tractor's pneumatic tires shall be mounted on steel, malleable iron, or cast iron wheels. Each wheel shall be mounted on wheel studs or cap screws with self-centering tapered ends or on tapered bolt holes which shall provide a positive method of locating the wheel on the hub.

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Wheels shall have demountable type rims which can be removed from hubs without deflating tires. Steer wheels shall clear any part of the tractor structure by at least 0.375 inch (10mm). Drive wheel clearance to any part of the tractor shall be sufficient to permit the use of standard automotive tire chains.

3.8.4 Tires.

3.8.4.1 Solid tires. When solid tires are specified (see 6.2), the tires shall be new and unused and shall be selected from sizes listed in the Tire and Rim Association Yearbook (T & RA). The tires shall be of the industrial solid rubber or cushion types molded onto a new press-on type steel band. All tires, both front and rear, shall have non-directional treads (smooth tires are not acceptable). Tire loadings shall not exceed the values shown in the Tire and Rim Association Handbook (T & RA) when tested in accordance with 4.6.3.

3.8.4.2 Pneumatic tires. When pneumatic tires are specified (see 6.2), the tires shall be new and unused and shall be selected from the sizes listed in the T&RA Yearbook. The tires shall be pneumatic, high pressure, tube type or tubeless, furnished with non-directional tread and smooth tires are unacceptable. Tire loadings shall not exceed the values listed in the T&RA Yearbook when tested in accordanced with 4.6.3.

3.8.5 Steering. An automotive type steering wheel, mounted on the steering column, shall turn the front wheels. Clockwise rotation of the steering wheel shall provide for right hand turning of the tractor while in forward motion. Provision shall be made for precision adjustment to compensate for wear of all major steering parts and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir. When provided the reservoir shall be accessible for filling or checking of lubricant level. Fifth wheel steer will not be acceptable. There shall be a minimum of 7 inches (17.8 cm) measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point on the unoccupied seat cushion. Power steering is acceptable.

3.8.6 Brakes.

3.8.6.1 Service brakes. The service brakes shall function without fade, noise, uneven braking or malfunction. The brake system shall be an automotive, four wheel, split system type, adequate for the size and weight of the tractor and load, and shall conform to the requirements of ANSI B56.9 (see 4.6.9). Split brake system shall provide means for stopping if either the front or rear brakes fail.

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3.8.6.2 Parking brake. The parking brake shall be an independent mechanical friction brake mounted on the drive motor or propeller shaft rotating with the wheels or a separate mechanical linkage actuating the service brake. The brake shall be automatically activated when the operator leaves the operator's seat. A hand operated parking brake lever with a locking device shall be provided in addition to the seat brake. This manual control shall be located within easy reach of the operator and in a position to permit easy and safe movement on and off the vehicle from either side. Adjustment of parking brake shoes or band shall be accomplished without the removal of any major assembly other than tires and wheels. The adjustment shall provide a means to prevent loss of adjustment through inadvertent disengagement of adjusting device. Unless otherwise specified (see 6.2) the seat brake will be required.

3.8.7 Body. The tractor body shall be constructed of steel sheet. Overall width shall be such that the tractor can perform the applicable tests in the Appendix of this specification. Provisions shall be made for immediate and easy access to all working parts requiring inspection or maintenance. Covers shall be secured with bolts or quick opening fasteners removable with ordinary hand tools.

3.8.7.1 Operator's compartment. The operator's compartment and seats shall be designed to accommodate a large sized, seated, clothed male and a passenger as specified by SAE J899.

3.8.7.2 Seats. Cushioned seats and cushioned backrests shall provide comfortable adjustable seating for the operator when operating the tractor and for the passenger. Unless otherwise specified (see 6.2), provisions shall be made to interrupt all power to the traction motor when the operator leaves the seat (see 3.7.2.5).

3.8.7.3 Towing hook. The tractor shall be provided with an automatic pintle-type hook. It shall automatically connect with trailers conforming to MIL-T-17479 and similar towing equipment.

3.8.7.4 Bumper. The tractor shall have a front bumper or bumper plate made of steel, and a rear bumper. Both the front and rear bumpers shall protect the tires from striking objects from the front and back of the tractor, respectively.

3.9 Controls and instrumentation. Where applicable, the provisions of UL 583 shall apply. Electric controls and instrumentation shall be constructed to be moisture and weather resistant. On type "EE" tractors, all instruments shall be of spark-enclosed construction or be mounted in a spark-proof enclosure.

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Controls and instrumentation locations and design shall conform to SAE J209/J297/J898.

3.9.1 Controls. All switch and lever type controls shall be provided with position markings conforming to SAE J1362 (decalcomania are not acceptable). Lever controls shall be equipped with hand grip knobs. All controls shall be easy to operate, positive in action, and shall show no evidence of deformation.

3.9.1.1 Light switch. An "on-off" light switch shall be located on the instrument panel for operation of the vehicle lights.

3.9.1.2 Brake pedal. The brake pedal shall be conveniently located for right foot operation within a clear space at least 6 inches (15.2 cm) wide. Pedal width shall be 3 inches (7.6 cm) minimum. Suitable means shall be provided for lubricating moving parts of the brake pedal and shaft. Required brake performamnce shall be attained with a pedal force of not more than 150 pounds (68kg).

3.9.1.3 Accelerator control. The accelerator control shall be installed for comfortable right foot operation. It shall be located to the right of, and shall be free of interference from the brake pedal by a distance of not less than 1.5 inches (3.8 cm).

3.9.1.4 Directional controls. Selective forward and reverse control lever(s) shall be provided and shall be conveniently located for right hand operation. Forward and reverse motion, however, shall be caused only by actuation of the accelerator pedal.

3.9.2 Instruments. All gages and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush in locations convenient to and plainly visible to the operator. All instruments shall be constructed to be moisture and weather resistant.

3.9.2.1 Hour meter. An electrically operationed hour meter, which registers the number of operating hours, shall be provided. The totalizing mechanism shall register not less than 9999. Hour meters for type 'E' and 'EE' trucks shall be in accordance with MIL-M-3971, Type I, Grade A. The meter shall be weather resistant. The hour meter shall be wired through the accelerator switch circuit to activate the meter when the accelerator switch is depressed. The hour meter shall be visible without opening compartments or removing panels.

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3.9.2.2 Battery Protector. Each unit shall be provided with a discharge indicator, of solid state construction, requiring no adjustments, shall be tamper-proof, weather and water resistant and shall carry a certified testing laboratory (eg., Underwriters Laboratories(UL) and Factory Mutual(FM)) approval. The unit shall display a warning light when the battery is 75% discharged and shall display a warning light and sound an alarm when the battery reaches 80% of discharge. Recharging or replacing the battery with one fully charged shall be the only means of resetting the warning lights and alarm.

3.10 Accessories. Where applicable, the provisions of UL 583 shall apply. All accessories shall be constructed to be moisture and weather resistant.

3.10.1 Headlights. Each tractor shall be equipped with 2 automotive type sealed beam headlights, of not less than 35 watts each. Headlights shall not protrude any further than the front bumper to prevent damage to the headlight. The headlights shall be vertically and horizontally adjustable.

3.10.2 Stoplight. One or more automotive, red, reflector type, combination stoplight and tail lamp, of not less than 15 candlepower and 3 candlepower respectively, shall be recessed or mounted on the rear end within the plan outline of the tractor. The stoplight shall operate automatically only upon actuation of the service brake pedal and shall not operate when the parking brake is applied.

3.10.3 Horn. Tractor shall be equipped with an electric horn with push button mounted on the center of the steering wheel or steering column. Horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant.

3.10.4 Closed cab. When specified (see 6.2), the tractor shall be equipped with a removable rigid closed cab conforming to paragraph 3.1 and Figure 1 of SAE J154. The cab shall include windshield, rear and side windows made of safety glass and shall provide maximum visibility. An electrically operated windshield wiper(s), a windshield defrosting unit,and a cab heater to maintain a minimum 65°F (18°C) shall also be provided.

3.10.5 Other Options. When specified (see 6.2), the tractor shall be equipped with fore and aft turn signal lamps and switch conforming to SAE J588f and SAE J589b, a rearview mirror conforming to SAE J985 (when a closed cab is required), side mirrors, or a backup alarm that conforms to SAE J994 Type D (87dB (A)).

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3.11 Slinging and tying down attachments. When specified(see 6.2), the tractors shall be equipped with lifting and tying down attachments conforming to MIL-STD-209, Type III. Attachments shall be indicated by a transportation plate conforming to MIL-P-514, except tying down attachments may be identified by stencilling or other suitable marking on the tractor. Tying down markings shall clearly indicate that the attachments are intended for tying down the tractor when shipped.

3.12 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubrication except for permanent type lubrication equipment. All lubricant reservoirs shall be appropriately filled with the correct lubricant.

3.12.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.12.2 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.13 Fungus and moisture resistance. The electric circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a moisture-proof type coating with the following exceptions:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

Before varnishing, all surfaces shall be cleaned and prepared so that they are free from all foreign matter which would interfere with the adherence or function of the varnish. The varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of .001 inch (.025 millimeters) to component or element surfaces.

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3.14 Identification marking. A shipping data plate shall be furnished and installed which shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material Type I, Grade A, Class I material and in addition, shall indicate the silhouette of the tractor in transport position indicating the center of gravity, the location and capacity of the lifting and the tiedown attachments and the contract number. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous protected locations.

3.14.1 Instruction plates. The tractor shall be equipped with instruction plates conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material, including warnings, cautions and diagrams describing any special or important procedures to be followed in assembling, operating or servicing the truck.

3.14.2 Wheel loading plate. Each tractor shall be equipped with a wheel loading plate conforming to MIL-P-514, Type III, Composition C, of Type 1, Grade A, Class 1 material. Plate shall be attached by screws, bolts, or rivets in a conspicuous location.

3.14.3 Battery plate. Each tractor shall be equipped with a battery information plate attached to the tractor conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material. As a minimum, the plate shall indicate battery weight, dimensions, ampere-hour capacity at a 6 hour rate, voltage, number of cells, number of plates, cable lead length and position.

3.15 Registration data and vehicle marking. The gross vehicle weight (GVW) including required battery shall be marked on each side of the tractor. The lettering shall be gothic capitals, and the numerals shall be 2 inches high where space permits. Tractors shall be marked in black color number 17038 in accordance with FED-STD-595. Marking paint shall be in accordance with TT-E-489. Marking location and information and character size shall be as follows:

Information	Example	Location	Character Size
Maximum Drawbar Pull	4,000 Lbs.	On each side	2 inch height
Service designation and Registration number	U.S. NAVY 11-XXXXX	On each side and rear	1-1/2 inch height

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Durable markers indicating the applicable safety designation of tractor shall be applied to each side of the vehicle in a visible but protected location.

3.16 Treatment and painting. All exterior surfaces of the tractor shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not,less than one coat of primer and two coats of synthetic enamel. Primer and finish coats shall be applied to assure complete coverage and durability of the finish. The finish color for the type E and EE trucks shall be color 13538 of FED-STD-595. The finish color for the type EX truck shall also be 13538, together with a single 6-inch (150 mm) wide blue diagonal stripe at 45 degrees across the right, left, and rear of truck; in addition, the type EX truck shall be identified as to type by painting 4-inch (100 mm) high black block letters with "EX" in a conspicuous space at right, left, and rear of truck. When a first produced truck is required for test and evaluation, it shall be painted with one primer coat only. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, peels, chips, blisters, areas of thin film or no film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with good commercial practice. Paint must be kept off working surfaces where interference with working of parts would result. The paint shall be lead free and chromate free. In addition, the truck shall be stenciled with the words "Painted with lead free, chromate free paint on (date) by (contractor's name)."

3.17 Repair parts and maintenance tools. All specially designed maintenance hand tools required for maintenance and repair of the tractors shall be furnished by the contractor.

3.18 Workmanship. The tractor shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the tractor before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.19 Human factors. The tractor shall conform to SAE J833 for the 5th and 95th percentile male and female.

3.20 Performance.

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3.20.1 Right angle turn. Tractors shall turn in 90 degree intersecting aisles with the widths listed below for each size tractor. Tractors shall turn in one motion, without backing, and shall meet this requirement in not more than 3 trials (see 4.6.5).

a. Size I - 70 in. (1.780 m)

b. Size II - 80 in. (2.032 m)

c. Size III - 90 in. (2.284 m)

d. Size IV - As specified

3.20.2 Steerability. The steer wheels shall be turned from extreme right to extreme left or vice versa in not more than 6 complete turns of the hand steering wheel. The force required to operate within the first half of the inner steer angle (from a straight ahead position) shall not exceed 40 pounds (177 Newtons), nor shall it exceed 80 pounds (355 Newtons) at the extreme ends of the steer in either direction when the tractor with two steer wheels is stationary on a smooth dry concrete floor. The hand steering wheel diameter shall not exceed 20 inches (510 mm). Other maneuverability features shall meet test course requirements Test Method No. 11 in the Appendix. The steering mechanism shall withstand an unbalanced tangential force of 125 pounds (555 Newtons) applied at the rim of the steering wheel without permanently distorting, fracturing or deforming any part of the steering mechanism, and at no time shall the free play be greater than 3 inches (76 mm) (see 4.6.6).

3.20.3 Speed. While pulling a load of 10 times the specified maximum drawbar pull on a level surface, the tractor shall attain at a minimum speed of 5 mph (8.0 km/hr) for Size I, 4 mph (6.4 km/hr) for Size II, and 3 mph (4.8km/hr) for Size III (see 4.6.7). The tractor shall be driven without trailing load on a level surface forward and reverse at a minimum speed of 6.5 mph (10.4 km/hr).

3.20.4 Ascending and Descending a Slope. Tractor shall ascend a 15 percent grade on a dry smooth concrete surface when pulling a load equal to the maximum drawbar pull(DBP) (i.e, a load of 40001b for a 40001b DBP). Tractors shall accelerate from a dead stop on this slope with trailing load (see 4.6.8).

3.20.5 Underclearance. The tractor shall have sufficient underclearance to permit operation from one horizontal plane to another, up or down a 15 percent slope, and on the obstacle course described Test Method No. 10 of the Appendix.

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Hydraulic fittings, hoses, tubing and linkages shall not extend below the lowest portion of the tractor and shall be protected by frame members from striking obstacles. (see 4.6.8)

3.20.6 Acceleration. From a standing start the tractor when pulling the rated load shall accelerate to a minimum of 6.5 mph (10.4 km/hr) in the first 22 feet (6.7 meters) (see 4.6.10).

3.20.7 Electromagnetic interference control. Type E and EE rated tractors shall comply with SAE J 551. When specified (see 6.2),type EE rated tractors shall be electromagnetically suppressed in accordance with class C1, Group 2, Part 8 of MIL-STD-461 (see 4.4.2.2).

3.20.8 Reliability. Tractor while pulling the rated load shall complete a minimum of 1200 laps in 100 hours of nonstop testing without failure when tested in accordance with 4.6.11. If failure occurs during the test, the cause of failure shall be corrected and the test restarted at the first lap.

3.20.9 Maintenance operations. Each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using common tools and special tools, if any, furnished with the tractor. (see 4.6.12)

Maintenance Operation (Time in Minutes)	E	EE
a. Remove and Replace Drive Motor Brushes	60	90
b. Remove and Replace All Contactor Tips	60	90
c. Remove and Replace All Filters, Screens and Strainers in Hydraulic System	105	105
d. Bleed and Adjust Brakes and Refill Master Cylinder (2 men)	45	45

3.20.10 Service brake. The brake system shall provide means for stopping the tractor if either the front or rear brakes fail to work. The brake pedal and brake system shall withstand a pressure of 250 pounds (1112 N) applied to the pedal without failure. The brakes shall hold the tractor stationary with a brake pedal pressure between 75 and 125 pounds (333 and 555 N) with an opposing force on the towing hook equal to the maximum drawbar pull (see 4.6.9).

3.20.11 Parking brake. The parking brake shall hold the tractor stationary (no tire movement is allowed) with a load equal to the maximum drawbar pull on a 15% slope in forward and reverse direction (see 4.6.8).

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3.20.12 Towing hook. The towing hook shall withstand, without physical failure, a towing force equal to the maximum drawbar pull induced in accordance with paragraph 4.6.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or purchase order, the contractor shall use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet the requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections.

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First Article Inspection. First Article inspection shall be performed on one complete tractor when a first article sample is required (see 3.1,6.2 and 6.5). This inspection shall include the examination of 4.5 and the tests marked "X" in the first column of Table II.

4.4 Quality Conformance Inspection. Quality conformance inspection shall include the examination of 4.5 and the tests marked "X" in the second column of Table II.

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4.5 Examination. First article examination shall include those items marked "X" in the first column of Table I. All tractors shall be examined as marked "X" in the second column of Table I. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5.1 Pretest examination Prior to testing in accordance with 4.6., the tractor(s) shall be examined for the defects as specified in Table I, Column I.

4.6 Tests. Upon successful completion of the pretest inspection, the tractor(s) shall be subjected to tests in accordance with Table II, Column 1. All tests shall be conducted in any order desired, except that the reliability and steering wear test shall be conducted last. Acceptance of the first produced tractor(s) shall not exclude the remaining tractors from the quality conformance inspections and tests specified in Section 4.

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TABLE I. Examination Schedule

First Article	Quality Conformance	Defect	Requirement Paragraph
x		Material not as specified	3.3
x	x	Casting not as specified	3.3.1
x		Plating not as specifed	3.3.2
x		Screw threads not as specified	3.3.3
x		Safety not as specified	3.4
x	x	Maintainability not as specified	3.6
x	x	Design not as specified	3.7
x	x	Enclosures and shields not as specified	3.7.1.2
x	x	Overcurrent devices not as specified	3.7.1.3
x		Heat producing devices not as specified	3.7.1.4
x	x	Static discharge devices not as specified	3.7.1.5
x		Control panel not as specified	3.7.1.6
x		Conductors not as specified	3.7.1.7
x	x	Motors and motor connections not as specified	3.7.2.1 3.7.2.2
x		Insulation not as specified	3.7.2.3
x		Speed controller not as specified	3.7.2.4
x		Safety cutout system not as specified	3.7.2.5
x	x	Battery not as specified	3.7.3
x		Battery connector mounting not as specified	3.7.3.1
x	x	Battery compartment not as specified	3.7.3.2
x	x	Drive unit not as specified	3.7.4
x	x	Chassis and frame not as specified	3.8.1
x		Bearings not as specified	3.8.2
x	x	Wheels not as specified	3.8.3
x	x	Tires not as specified	3.8.4
x		Steering not as specified	3.8.5
x		Service brake not as specified	3.8.6.1
x	x	Parking brake not as specified	3.8.6.2

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TABLE I. Examination Schedule(con't)

First Article	Quality Conformance	Defect	Requirement Paragraph
x		Tractor body not as specified	3.8.7
x	x	Operators compartment not	3.8.7.1
x	x	Seat not as specified	3.8.7.2
x	x	Towing hook not as specified	3.8.7.3
x	x	Bumper not as specified	3.8.7.4
x	x	Controls and instrumentation not as specified	3.9
x	x	Battery protector not as specified	3.9.2.2
x	x	Headlights not as specified	3.10.1
x	x	Stoplight not as specified	3.10.2
x	x	Horn not as specified	3.10.3
x	x	Slinging and tiedown attachments not as specified	3.11
x	x	Lubrication and fittings not as specified	3.12.1
x	x	Fungus and moisture resistance not as specified	3.13
x	x	Identification markings not as specified	3.14
x	x	Instruction plate not as specified	3.14.1
x	x	Wheel loading plate not as specified	3.14.2
x	x	Battery plate not as specified	3.14.3
x	x	Markings not as specified	3.15
x	x	Treatment and painting not as specified	3.16
x		Human factors not as as specified	3.19

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TABLE II. Test Schedule

First Article	Quality Conform.	Test	Test Para.	Requir. Para.	Test Method No.
x		Steering wheel test	4.6.6	3.20.2	1
x		Wear of steer mechanism	4.6.6	3.20.2	2
x	x	Speed	4.6.7	3.20.3	3
x	x	Ascending a slope		3.20.4	4
		Underclearance,		3.20.5	
		Parking brake,		3.20.11	
		Ascending a slope	4.6.8	3.20.4	
x		Service brakes	4.6.9	3.20.10	5 & 6
x		Right angle turn	4.6.5	3.20.1	7
x		Acceleration	4.6.10	3.20.6	8
x		Towing hook	4.6.4	3.20.12	9
x		Reliability test	4.6.11	3.20.8	10
x		Maintainability	4.6.12	3.20.9	11
x	x	Run-in test,	4.6.1	Sect. 3	
x		Electromagnetic interference	4.6.2	3.20.7	

4.6.1 Run-in. Pulling a trailing load of 10 times the maximum DBP, the tractor shall be run-in for not less than 30 minutes on a level circular course having a diameter not exceeding 100 feet (25.4 meters). The speed for this 30 minute period shall average not less than 5 miles per hour (8.0 kilometers per hour). Three laps shall be conducted at a speed not less than 6.5 miles per hour (10.4 kilometers per hour). During operation and after completion of run-in, the tractor shall be examined for leaks, operational failures, failure of any component to perform as specified, or permanent deformation. Any evidence of the above shall constitute failure of the tests.

4.6.2 Electromagnetic interference. Test to determine compliance with 3.20.7 shall be performed in accordance with the requirements of SAE J551 or MIL-STD-461 (see 6.2). Failure to meet requirements will be cause for rejection.

4.6.3 Tires. With a scale determine the load on each tire. Tires must conform to the requirements of 3.8.4.

4.6.4 Towing hook. Test tractor in accordance with Test Method No. 9. Nonconformance to 3.8.7.3, any malfunction, overheating, leakage of fluid, or failure of any components shall be cause for rejection.

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4.6.5 Right angle turn. Test tractor in accordance with Test Method No. 7. Failure to conform with the requirements of 3.20.1 shall constitute failure of the test.

4.6.6 Steering. Test tractor in accordance with Test Methods No. 1 and 2. Nonconformance with the requirements of 3.20.2 shall constitute failure of the test.

4.6.7 Speed. Test the tractor in accordance with Test Method No. 3. Noncompliance with 3.20.3 shall constitute failure of the test.

4.6.8 Ascending a Slope, Parking brake, Underclearance, Descending a Slope. Test the tractor in accordance with Test Method No. 4. Nonconformance to 3.20.4, 3.20.5 or 3.20.11 shall constitute failure of this test.

4.6.9 Service brakes. Test the tractor in accordance with Test Method No. 5 and 6. Nonconformance with 3.20.10 shall constitute failure of this test.

4.6.10 Acceleration. Test the tractor in accordance with Test Method No. 8. Nonconformance to 3.20.6 shall constitute failure of this test.

4.6.11 Reliability. Test the tractor in accordance with Test Method No. 10 (see 3.20.8). Failure for the purpose of the test is defined as any malfunction which cannot be remedied by adjustment, repair or replacement by one person using the organizations tools and parts within 15 minutes (whether or not parts are available) and which may cause:

a. Failure to start operation, stop of operation, or degradation of performance capabilities of the tractor below designated levels.

b. Serious damage to the tractor by continuous operation.

c. Serious personnel safety hazard.

Delays in testing over 15 minutes for any reason are unacceptable. The reliability testing must be restarted at the first lap for such delays. When the total time of delays equals one hour, the test shall be restarted from the initial lap.

4.6.12 Maintainability. Test the tractor in accordance with Test Method No. 11. Nonconformance to 3.20.9 shall constitute failure of this test.

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4.6.13 Failure. Inability to meet or noncompliance with any of the test requirements will constitute failure. Each test failed must be restarted at the beginning of that test. Modification of any components of the tractor during testing shall be cause for restarting the entire first article testing.

4.7 Quality conformance inspection.

4.7.1 Tests. Each tractor shall be subjected to the tests in accordance with Table II, column 2. Nonconformance with applicable requirements shall constitute failure. In the event of failure dismantle the tractor sufficiently to permit detailed disassembly and visual inspection of any faulty components.

4.7.2 Inspection. After successful completion of all tests specified in 4.6, each tractor shall be examined in accordance with Table I, column 2. Presence of one or more defects shall be cause for rejection.

4.8 Inspection comparison. The Government may select tractors at any time during the contract production period to be inspected as specified in column 1 of Table I or column 1 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, and will not include the previously inspected first produced tractor(s). In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison. Failure of such additional tests shall have the same effect as failure of the tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison tractor to meet any requirement specified herein during and as a result of the inspection and tests specified in 4.6 shall be cause for rejection of the inspection comparison tractor(s) and shall be cause for refusal by the Government to continue acceptance of the production tractors until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiences. Correction of the deficiences shall be accomplished by the supplier at no cost to the Government on tractors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all tractors accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.9 Inspection of preparation for delivery. An inspection will be made to determine that the requirements for preservation, packaging, packing and marking are in compliance with 5.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation, packaging and packing shall be in accordance with MIL-STD-162 or ASTM D3951 as specified.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The four-wheeled, sit-down, electric-motor driven, warehouse tractor is to be used for pulling trailers loaded with cargo in and around warehouses, loading platforms and docks within the military supply system. Tractors are intended for operation over paved and improved surfaces.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.

(b) Type of tractor required (see 1.2.1).

(c) Maximum drawbar pull required (see 1.2.2).

(d) Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1).

(e) When first article tractor(s) is (are) required (see 3.3).

(f) When the Government will conduct any or all of the first produced tests (see 3.1).

(g) When plating shall be for shipboard or saline atmosphere (see 3.3.2).

(h) When battery size and capacity are not according to MS-15367 and W-B-133 (see 3.7.3).

(i) When seat switch is not required (see 3.7.2.5.1, 3.7.2.5.2, 3.8.6.2 and 3.8.7.2).

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(j) When single point watering system is required (see 3.7.3).

(k) Required type of tires and wheels (see 3.8.3 and 3.8.4)

(l) When rigid closed cab is required (see 3.10.4).

(m) When other options are required (see 3.10.5).

(n) When slinging and tiedown attachments are other than specified (see 3.11)

(o) When paint shall be other than required (see 3.16).

(p) Required test method for electromagnetic interference control (see 3.20.7).

(q) Level of preservation and packing required (see 5.1).

(r) Special markings if required in addition to MIL-STD-129 (see 3.15).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To insure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2	DI-T-4902	First Article Inspection Test Report	-----

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management System and Data Requirements Control List (AMSDL), must be researched to insure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical Manuals. The requirements for technical manuals should be considered when this specification is applied on a contract.

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If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.3. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Conditions for use of level B preservation. When level B preservation is specified (see 5.1), this level of protection should be reserved for the acquisition of tractors for resupply worldwide under known favorable handling, transportation and storage conditions.

6.7 Subject term (key word) listing.

Tractor
Warehouse
Trailer
Maximum drawbar pull
Towing hook
Battery

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - ME
Navy - SA
Air Force - 99

Preparing Activity:

Navy - SA

Project No. 3930-0672

Review Activites:

Navy - MC, YD,
Air Force - 84
DLA - CS
Army - MT, A

User Activities:

Navy - OS
DLA - DOSO-DER

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APPENDIX

FOR

TRUCK, FIXED PLATFORM, ELECTRIC POWERED

10. SCOPE

The test methods contained within this appendix are to determine whether trucks procured under this specification conform to the requirements set forth. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. TEST METHODS

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TEST METHOD NO. 1

STEERING WHEEL TEST

Purpose: To determine whether the force required to turn the steering wheel meets specification.

Test Description: Place the tractor on a level concrete surface. Using a tension dynamometer or equivalent, measure the force required to move the steering wheel.

Test Procedure:

1. Place the tractor on dry, level, smooth, trowelled or brushed concrete with steer wheel(s) in a straight-ahead position.

2. Assemble an adapter pulley having a diameter equal to the steering wheel diameter and equipped with means for attachment to the wheel spokes.

3. Attach a tension dynamometer to the adapter pulley.

4. Grasp the dynamometer lead and apply a steady force, in a counter-clockwise direction and in the wheel plane, until the wheel turns.

5. Observe the maximum gauge reading.

6. Perform 4 and 5 in a clockwise direction.

7. Turn steering wheel clockwise until the steer wheels hit the stop. Apply a 125lb unbalanced tangential force to the steering wheel and maintain the force for a minimum of 15 seconds. Perform this operation two more times.

8. Repeat step 7 in the counter-clockwise direction.

9. Measure the number of turns of the hand steering wheel required to turn the steer wheels from the extreme right position to the extreme left position. Repeat this operation in the opposite direction.

10. Measure the steering wheel diameter, and the steering wheel to seat clearance.

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TEST METHOD NO. 2

WEAR OF STEERING MECHANISM *

Purpose: To determine cumulative wear in the steering mechanism of the tractor due to carrying out requirements of the reliability test.

Test Description: With the tractor stationary, the amount of free play in the steering wheel is measured.

Test Procedure:

1. Place the tractor on a dry, level, concrete surface with its front wheels parallel to the longitudinal axis.

2. Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semi-circular in shape and which has a radius of at least one inch greater than the radius of the hand steering wheel.

3. Mark a reference point on the upper portion of the outer edge of the hand steering wheel; mark a corresponding point on the fiber board template.

4. Turn the steering wheel counterclockwise until the front wheels of the tractor begin to turn. Mark a point in the fiber board template corresponding to the reference point on the outer edge of the steering wheel.

5. Turn the steering wheel clockwise until the front wheels of the tractor begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

6. Remove fiberboard templet from tractor.

7. Place fiberboard templet on a flat surface and measure the distance between two points made in accordance with operation 4 and 5 above. This distance, in inches, is the free play of the steering mechanism.

*This test should be done before and after reliability test.

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TEST METHOD NO. 3

SPEED

Purpose: To determine whether the tractor meets the speed requirements of this specification.

Test Description: Measuring the speed of the the tractor while pulling a trailing load of 10 times the maximum DBP (rated load), and without the rated load in forward and reverse, a distance of at least 88 feet (27 meters).

Test Procudure:

1. Drive the tractor a sufficient distance to attain maximum speed prior to entering the measured test course.

2. Record the length of measured distance and time to transverse the measured distance.

3. Repeat for a total of six runs, in forward gear, three in each direction.

4. Repeat a total of six runs in forward and reverse gear without trailing load, three in each direction.

5. Travel speeds in forward and reverse shall be the average of the six runs.

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TEST METHOD NO. 4

ASCENDING A SLOPE-PARKING BRAKE - UNDERCLEARANCE -

DESCENDING A SLOPE

Purpose: To determine whether the tractor, when pulling a trailing load, can accelerate from a dead stop on a 15 percent grade and whether it has sufficient underclearance to operate from a 15 percent slope to a horizontal surface. To determine whether the parking brake can hold tractor with trailing load on a 15 percent slope in both the upward and downward directions.

Test Description: Tractor ascends a 15 percent grade in a forward direction on a dry concrete surface. Tractor stops and starts up slope again, then descends the ramp. Horizontal approaches should be provided for the ramp.

Test Procedure:

1. Place a load equal to the DBP on trailers.

2. Drive tractor along horizontal approach to the 15 percent ramp.

3. Drive tractor approximately half-way up the 15 percent ramp and bring to a complete stop.

4. Apply parking brake, release service brakes and observe whether tractor remains stationary.

5. Release parking brake, and start tractor from dead start and proceed up ramp and onto a horizontal surface. Observe underclearance.

6. Descend ramp. Stop on ramp and apply parking brakes, release service brakes, and observe whether tractor remains stationary.

7. Proceed on to the horizontal surface.

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TEST METHOD NO. 5

SERVICE BRAKE PRESSURE

Purpose: To determine whether brake pedal and brake system have the strength required by the specifications.

Test Description: Using a pedal force gauge, apply a pressure of 250 pounds (1112 Newtons) for one minute to the brake pedal a total of 10 times.

Test Procedure:

1. Attach the pressure gauge to the brake pedal in a manner which enables pressure to be applied to the face of the gauge.

2. Apply foot pressure to the gauge until it registers 250 pounds (1112 Newtons) and hold for 1 minute.

3. Repeat operation number two, 10 times.

4. Examine systems for any leaks and structural deformation, or broken welds.

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TEST METHOD NO. 6

SERVICE BRAKE PRESSURE SENSITIVITY,

SERVICE BRAKE STRENGTH

Purpose: To determine whether a brake pedal pressure within the range of 75 to 125 pounds (333 to 555 Newtons) will withstand the maximum drawbar pull.

Test Description: Pull tractor over dry, smooth, trowelled or brushed concrete with parking brake released.

Test Procedure:

1. Attach pressure gauge to brake pedal, in a manner which will enable pressure to be applied to the face of the gauge.

2. Attach a tension dynamometer to the front of the tractor.

3. Attach one end of a block and tackle, or similar device for securing mechanical advantage, to the tension dynamometer, and the other end to an anchoring device.

4. Apply sufficient brake pedal pressure to prevent wheel from turning.

5. Apply tension to the dynamometer until force is approximate equal to maximum DBP.

6. Gradually reduce brake pedal pressure until wheels just begin to turn while maintaining the dynamometer tension. This pedal pressure shall not be less than 75 pounds (333 Newtons), nor more than 125 pounds (533 Newtons).

7. Repeat operations 1 through 6 with dynamometer attached to rear towing hook.

8. Examine towing hook for evidence of permanent deformation.

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TEST METHOD NO. 7

RIGHT ANGLE TURN

Purpose: To determine whether tractor can turn in 90 degree intersecting aisle of specified width.

Test Description: 90 degree intersecting aisles of specified width are laid out. The tractor must make a right hand turn within the specified aisle widths in one motion, without backing, and with no more than three trials. The test is repeated with the tractor making left hand turns.

Test Procedure:

1. Layout on dry, smooth concrete two 90 degree intersecting aisles of specified width.

2. Drive tractor at slow speed, in one motion and without backing, around the intersecting aisles to the right. Tractor must negotiate the turn within the specified aisle width in not more than three trials.

3. Perform operation 2 with tractor turning to the left.

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TEST METHOD NO. 8

ACCELERATION

Purpose: To determine acceleration of tractor pulling the rated load (see test method No. 3) on a level surface from a standing start.

Test Description: Tractor is accelerated at maximum rate from a standing start, timed through a distance of 22 feet (6.7 meter), and then brought to a stop.

Course Layout:

A----------22' (6.7 m)---------B

Test Procedure:

1. Accelerate the tractor in forward gear at maximum speed through the specified distance from a standing stop.

2. Repeat a total of six runs, three in each direction.

3. Final results shall be the average of the six runs.

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TEST METHOD NO. 9

TOWING HOOK TEST

Purpose: To determine whether the tractor towing hook can withstand specified maximum drawbar pull without permanent visible deflection or deformation.

Test Description: With tractor on a level surface, specified drawbar pull shall be exerted on the tractor pintle hook. There shall be no permanent deformation or failures of related structures.

Test Procedure:

1. Place the tractor on a level, clean, dry, smooth concrete or paved surface.

2. Attach a tension dynamometer between a fixed anchor and the tractor's pintle installed 12 inches, plus or minus 0.5 inch (12 mm) above ground level.

3. Operate the tractor at maximum drawbar pull for at least 30 seconds and record drawbar pull at 5 second intervals.

4. If the wheels slip during testing, that trial must be performed again.

5. Repeat at least five times.

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TEST METHOD NO. 10

RELIABILITY TEST

1. Test Course:

(a) Layout of course. The test course shall be set up in accordance with the provisions of Figure 1. The length of each leg may be varied, so long as the total length of the course and its general proportions are maintained. Aisle or road widths shall be 12 feet (3.6 meters). The perimeter of the course shall be encompassed with suitable barriers or indicators. Rubber pylons, or other suitable markings shall be placed at all corners and other critical points to assist in keeping the machine under test within the course limits.

(b) Ramp inclusion. The course shall be laid out to include the slope (plus or minus one percent) and approximate length specified on Figure 3.

(c) Obstacle inclusion. Obstacles shall be included in the course layout as indicated on Figure 2. Iron, wood, cement or steel blocks may be used. A guideline 15 feet (4.6 meters) long shall be provided to the left of the centerline of the course and parallel to it. This line shall be at a distance from the course centerline equal to one-half the distance between front wheel centers when two steer wheels are used. This spacing and arrangement will allow the left wheels of the machine under test to pass over the first pair of blocks and then the right wheels to pass over the second pair of blocks while the machine is moving forward in a straight line with the left front wheel tracking on the guideline. The tractor shall transverse the obstacle course on every tenth lap.

(d) Course surface. The test course shall be concrete or macadom, and shall be free of snow, ice, any non-planned obstacles or foreign materials during the course of test.

(e) Minimum lighting requirements. Minimum general illumination of the course shall be 5-foot-candles (5.38 lux) on the operating level. However, lights of the tractor shall be on at all times during the test. The light switch shall be turned "off" and "on" at the beginning of each lap.

(f) Changing batteries. Each machine under test shall utilize not more than two sets of fully charged batteries during the course of each operating 8-hour day of the performance test.

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However, the time required to change batteries under the provisions of this paragraph shall be excluded from the operating time. Batteries used shall have minimum ratings as specified. No other adjustments taking longer than 15 minutes may be made at this time .

(g) Speed of negotiating course. Each machine undergoing test should tranverse each part of the performance test course at the maximum safe practical speed.

(h) Hours per day. Each tractor shall complete a minimum of 1200 complete circuits of the test course in 100 hours of testing. Circuits (laps) shall be conducted on a continuous basis during each day (8 hour minimum).

(i) Duration of the performance test. The performance test specified in this standard shall be conducted on a continuous basis each operating day until the minimum requirements are accomplished.

(j) Periodic maintenance. On the fourth operating day (after 24 hours) of the performance test the truck shall undergo routine maintenance procedures. However, no repairs or replacement of parts shall be permitted.

(k) Method of negotitating course. The tractor with a trailing load of 10 times the maximum drawbar pull shall begin each lap at the point marked 'START' on Figure 1. On every lap, the tractor shall climb and descend the five percent ramp. On every tenth lap, the tractor must stop halfway up the ramp, point 'A' on Figure 1, then resume completing the course. On the very same lap, the tractor shall transverse the obstacle course as shown on Figure 2. Stopping before the obstacle course is not permitted. After each eight (8) hours of testing, the tractor shall transverse the course in the opposite direction. The " Hourly Record Sheet " shall be completed for each operating hour of performance testing.

(l) Operating limitations. The tractor under test shall not be permitted to leave the bounds circumscribed in Figure 1.

(m) Failure. Any failure of the tractor during the performance testing that requires more than fifteen minutes to repair (one person) shall be cause to restart the test from the first lap.

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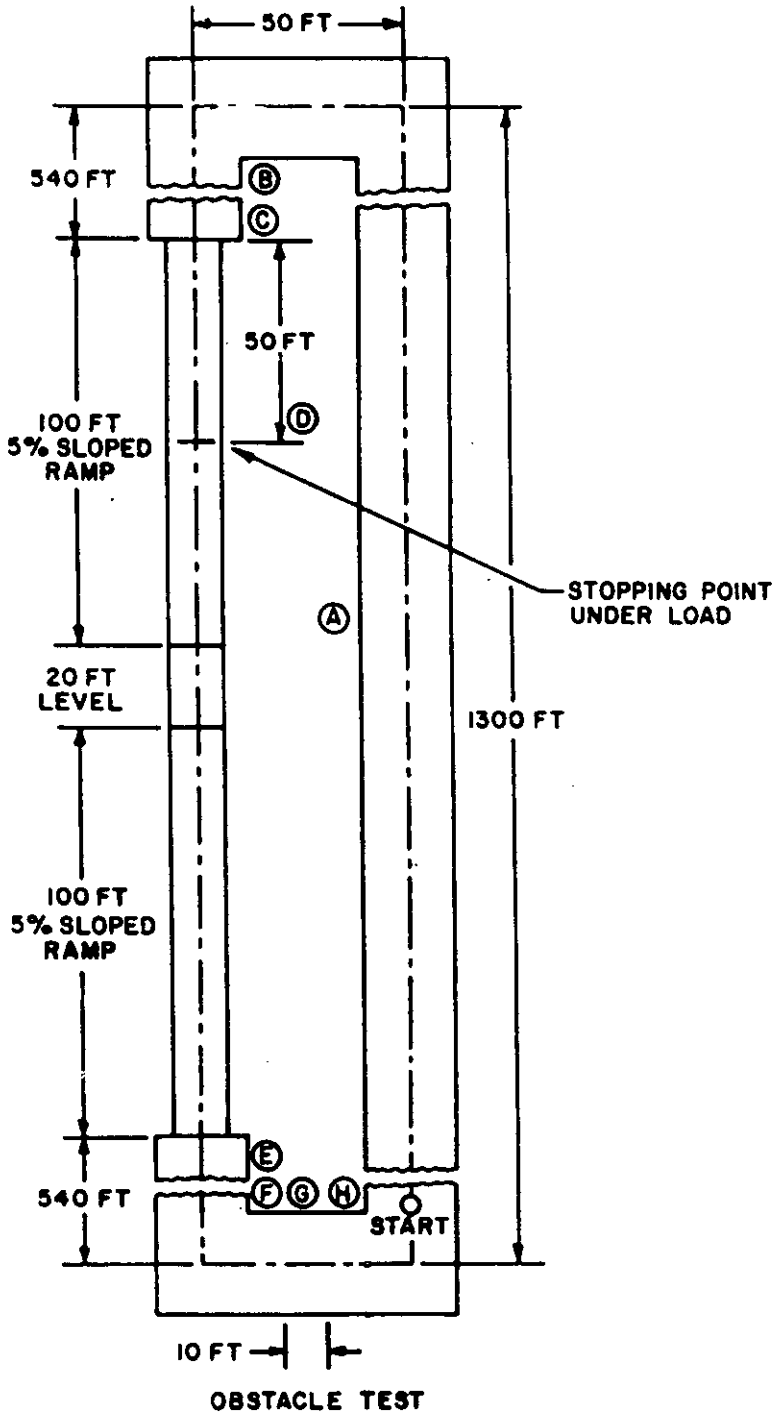
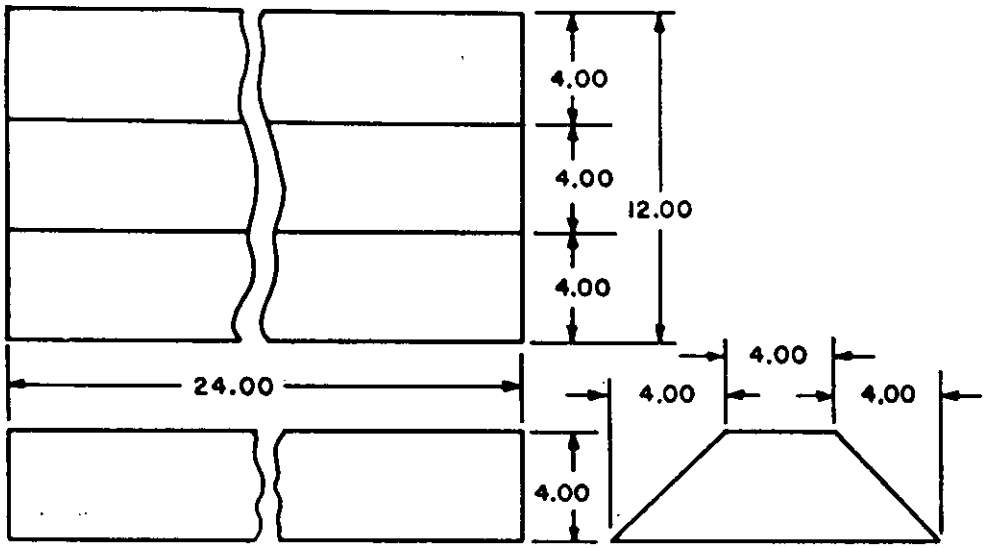


FIGURE 1. Test course for tractors.

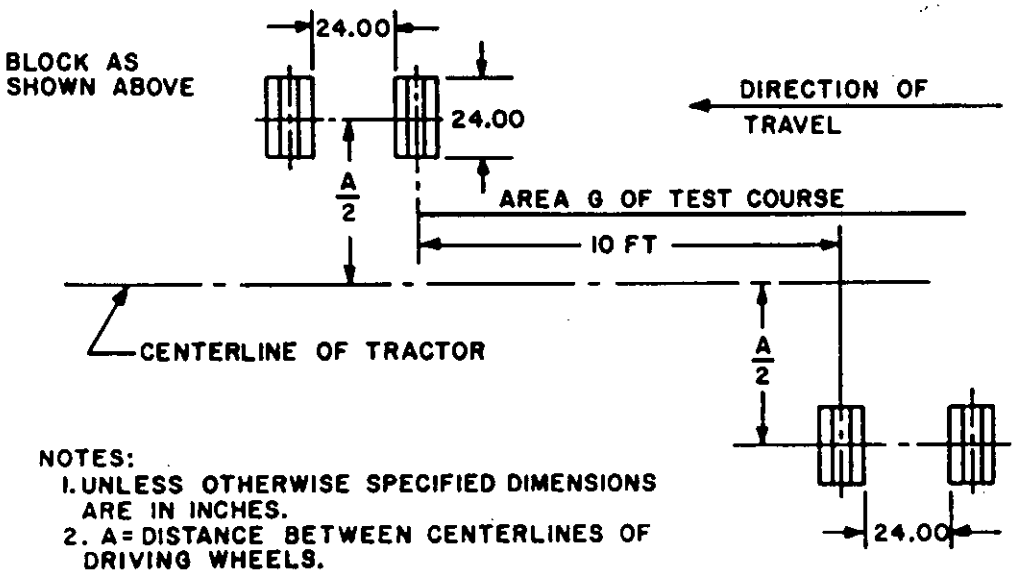
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OBSTACLE BLOCK CONSTRUCTION



LAYOUT OF OBSTACLE COURSE

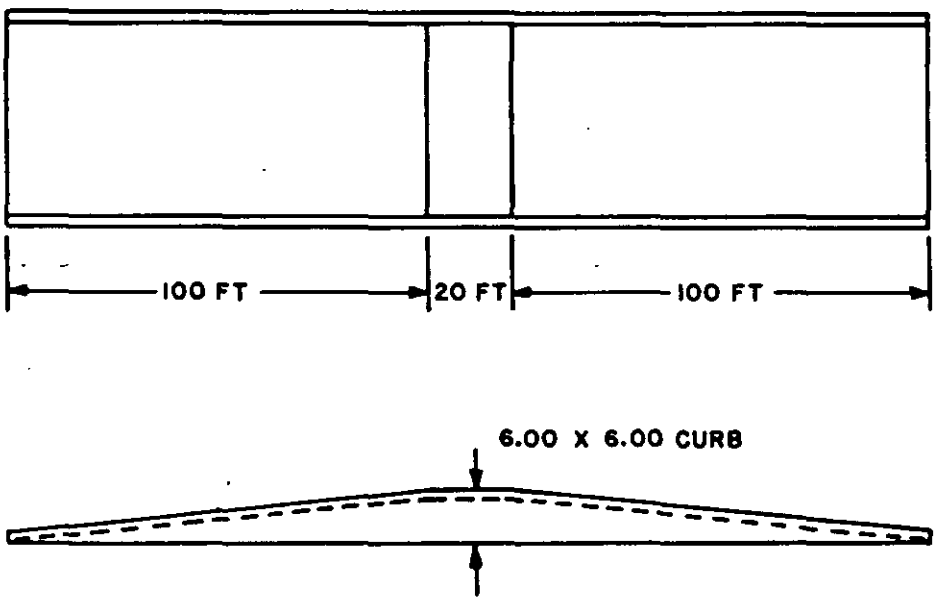


NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. A = DISTANCE BETWEEN CENTERLINES OF DRIVING WHEELS.

FIGURE 2. Obstacle block layout.

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NOTE: UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

FIGURE 3. Test ramp.

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HOURLY TIME RECORD

TRUCK MANUFACTURER	MODEL	SERIAL NO.	DATE

TEST LOCATION	RECORDED BY	CLOCK READINGS:	(Start)	(Finish)	REQ'D LAPS/HR

LAP NUMBER	WATCH READING	ELASPED TIME	REMARKS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
TOTAL			AVERAGE LAP TIME

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Instructions for Timing Performance Test

1. Start decimal minute stop at zero at beginning of each operating hour.

2. Record watch reading at end of each lap in "Watch Reading" space.

3. Subtract watch reading at end of preceding lap from reading at end of lap. Record difference in "Elapsed Time" space.

4. If equipment under test completes the required number of laps prior to completion of an operating hour, it shall be stopped until start of next hour, and then resume performance test.

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TEST METHOD NO. 11

MAINTAINABILITY

1. Test Course:

Shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. Test Apparatus:

(a) Common handtools
(b) Special tools (if furnished with tractor)

3. Test Procedure:

(a) Examine tractor for accessibility of major assemblies.
(b) Examine position of drains with respect to accessibility.
(c) Using the tools, demonstrate that major assemblies are accessible for repair and maintenance.
(d) Demonstrate both the accessibility of drains and the path of discharge of lubricants by activating drainage controls.
(e) Perform and time each operation on the maintenance operations list (see 3.20.9).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-10010J	940628

3. DOCUMENT TITLE
TRACTOR, FOUR WHEELED, SIT DOWN, WAREHOUSE, ELECTRIC-MOTOR DRIVEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDING OFFICER ATTN: CODE 0341 (M. MYERS)	(1) Commercial (717) 790-4024	(2) AUTOVON 430-4024
c. ADDRESS (Include Zip Code) NAVY SHIPS PARTS CONTROL CENTER P.O. BOX 2020, 5450 CARLISLE PIKE MECHANICSBURG, PA 17055-0788	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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